Threshold Analysis of the Rotated Surface Code Under Depolarizing Noise: A Comparative Study of MWPM and Neural Network Decoders

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March 2026

Abstract

Quantum error correction (QEC) is a prerequisite for fault-tolerant quantum computation. This paper presents a Monte Carlo simulation of the rotated surface code under a depolarizing noise model across code distances $d \in \{3, 5, 7, 9, 11\}$, implemented using the Stim stabilizer simulator. Two decoders are compared: minimum-weight perfect matching (MWPM) via PyMatching, and a feedforward neural network (NN) trained on syndrome data. A threshold of approximately $p_{\text{th}} \approx 0.01$ is identified for MWPM, consistent with theoretical predictions. The NN decoder performs significantly worse than MWPM at low error rates due to class imbalance in the training data, motivating future work on improved learned decoders. Below threshold, MWPM achieves exponential logical error suppression with increasing code distance.

1 Introduction

Physical qubits are subject to decoherence, gate imperfections, and environmental noise. Without error correction, quantum computations cannot proceed beyond a small number of operations. Quantum error correcting codes protect logical information by distributing it across multiple physical qubits such that errors can be detected and corrected without measuring the encoded state directly.

The surface code [1] is the leading candidate for fault-tolerant quantum computation due to its high threshold, local stabilizer measurements, and compatibility with planar qubit architectures. The rotated surface code [2] reduces qubit overhead while preserving error correction properties.

This paper addresses two questions: (1) how does logical error suppression scale with code distance from $d = 3$ to $d = 11$ under MWPM decoding, and (2) can a simple feedforward neural network match MWPM performance as a decoder?

2 Background

2.1 Qubits and Pauli Errors

A qubit exists in superposition $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ where $|\alpha|^2 + |\beta|^2 = 1$. Three independent error types act on qubits:

$$X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle \quad \text{(bit-flip)} \tag{1}$$

$$Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle \quad \text{(phase-flip)} \tag{2}$$

$$Y = iXZ \quad \text{(combined bit and phase flip)} \tag{3}$$

2.2 Stabilizer Formalism

A stabilizer code is defined by commuting Pauli operators $\{S_i\}$ whose simultaneous $+1$ eigenspace defines the code space. When error E anticommutes with stabilizer S_i:

$$S_i(E|\psi\rangle) = -(E|\psi\rangle) \tag{4}$$

The eigenvalue flips to -1, yielding a classical syndrome bit identifying the error without collapsing the encoded quantum state.

2.3 The Rotated Surface Code

A distance-d rotated surface code encodes one logical qubit in d^2 data qubits with $(d^2-1)/2$ X-stabilizers and $(d^2-1)/2$ Z-stabilizers. The code distance d is the minimum weight of an undetectable logical error.

Table 1: Qubit counts for each code distance

Distance d	Data Qubits	Total Qubits (with ancilla)
3	9	26
5	25	58
7	49	106
9	81	170
11	121	250

3 Methodology

3.1 Noise Model

A symmetric depolarizing channel was applied with physical error rate p:

$$\mathcal{E}(\rho) = (1-p)\rho + \frac{p}{3}(X\rho X + Y\rho Y + Z\rho Z) \tag{5}$$

Two-qubit gates receive a depolarizing channel with per-Pauli probability $p/15$.

3.2 MWPM Decoder

Circuits were generated using Stim [5] with the `surface_code:rotated_memory_z` template. Decoding was performed using PyMatching [6] via minimum-weight perfect matching on the detector error model. Each configuration was sampled with $N = 100{,}000$ Monte Carlo shots across 15 logarithmically spaced error rates from $p = 10^{-3}$ to $p = 10^{-1}$.

3.3 Neural Network Decoder

A feedforward neural network was trained to classify syndromes as logical error or no logical error. Architecture: input layer (syndrome size) $\rightarrow$ 128 units (ReLU) $\rightarrow$ 64 units (ReLU) $\rightarrow$ 1 unit (Sigmoid). Training used 30,000 syndrome-label pairs per configuration, Adam optimizer, binary cross-entropy loss, 15 epochs. Evaluation used 10,000 held-out shots.

Table 2: Simulation parameters

Parameter	Value
Code distances (MWPM)	$d \in \{3, 5, 7, 9, 11\}$
Code distances (NN)	$d \in \{3, 5, 7\}$
Error correction rounds	10
MWPM shots	100,000 per configuration
NN training shots	30,000 per configuration
NN evaluation shots	10,000 per configuration
Physical error rate range	$[10^{-3}, 10^{-1}]$
Simulator	Stim v1.15.0

4 Results

4.1 MWPM Threshold Analysis

Figure 1 shows logical error rates for $d \in \{3, 5, 7, 9, 11\}$. All curves converge near $p_{\text{th}} \approx 0.01$, confirming the threshold.

4.2 Below-Threshold Scaling

At $p = 0.001$, logical error rates under MWPM were:

Table 3: MWPM logical error rates at $p = 0.001$

Distance d	p_L	Suppression vs $d = 3$
3	0.00231	$1\times$
5	0.00027	$8.6\times$
7	0.00003	$77\times$
9	0.00000	$> 231\times$
11	0.00000	$> 231\times$

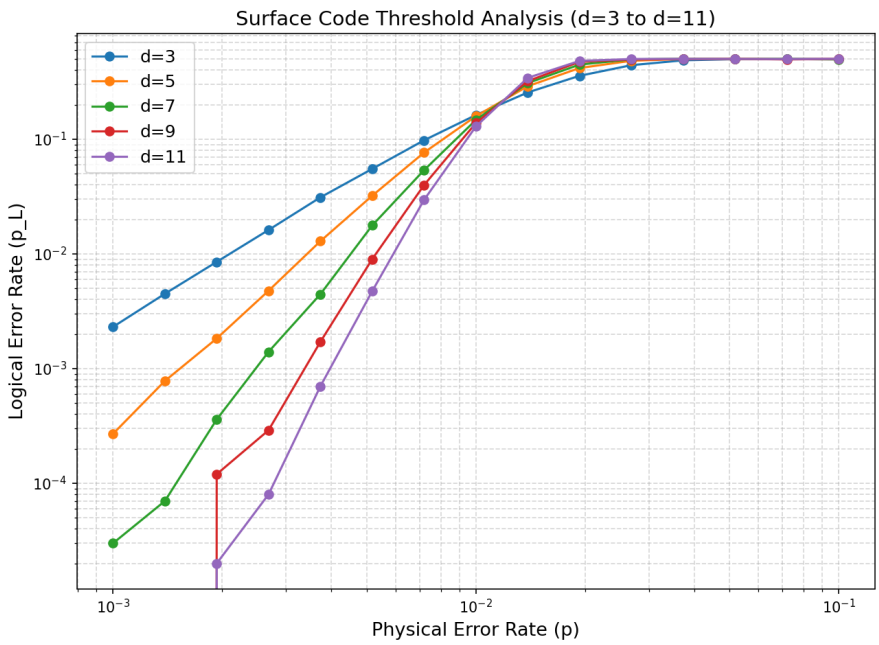

Figure 1: Logical error rate vs physical error rate for $d \in \{3, 5, 7, 9, 11\}$ under MWPM decoding. Curves converge near $p \approx 0.01$.

Each distance increment of 2 reduces the logical error rate by approximately one order of magnitude, consistent with the theoretical scaling:

$$p_L \sim \left(\frac{p}{p_{\text{th}}}\right)^{\lfloor (d+1)/2 \rfloor} \tag{6}$$

4.3 MWPM vs Neural Network Decoder

Figure 2 compares MWPM and NN decoder performance for $d \in \{3, 5, 7\}$.

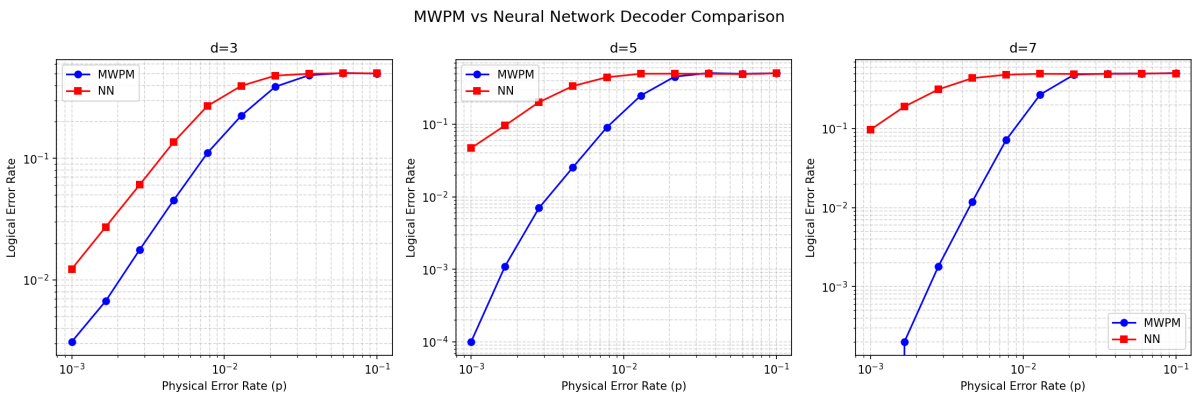

Figure 2: MWPM vs neural network decoder logical error rates for $d = 3, 5, 7$. MWPM consistently outperforms the NN decoder, particularly at low error rates.

At $p = 0.001$ for $d = 7$, MWPM achieves $p_L \approx 0$ while the NN decoder produces $p_L \approx 0.097$. This gap arises from class imbalance: at low physical error rates, logical

errors are rare, providing insufficient positive training examples for the network to learn the error structure. The NN decoder converges toward random guessing ($p_L \approx 0.5$) at high error rates, while MWPM degrades more gracefully.

Table 4: MWPM vs NN logical error rate at $p = 0.001$

Distance d	MWPM p_L	NN p_L	Ratio
3	0.00310	0.01230	4×
5	0.00010	0.04690	469×
7	0.00000	0.09700	>1000×

5 Discussion

The NN decoder's failure at low error rates is a known limitation of supervised learning approaches applied to QEC [7]. Mitigation strategies include data augmentation, importance sampling of rare error events, and recurrent architectures that exploit temporal syndrome correlations across rounds. These remain open research directions.

MWPM remains optimal for depolarizing noise on the surface code because the detector error model maps exactly to a minimum-weight matching problem. Learned decoders may outperform MWPM under non-Pauli or correlated noise models where the MWPM graph construction becomes approximate.

6 Conclusion

This study confirms the threshold theorem for the rotated surface code under depolarizing noise. MWPM decoding achieves a threshold of $p_{\text{th}} \approx 0.01$, with exponential logical error suppression below threshold as distance increases from $d = 3$ to $d = 11$. A feedforward neural network decoder was found to perform significantly worse than MWPM at low error rates due to class imbalance, highlighting the challenge of applying standard supervised learning to QEC decoding.

Future work will investigate recurrent neural networks, transformer-based decoders, and belief propagation approaches, with evaluation under circuit-level noise models beyond depolarizing.

References

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